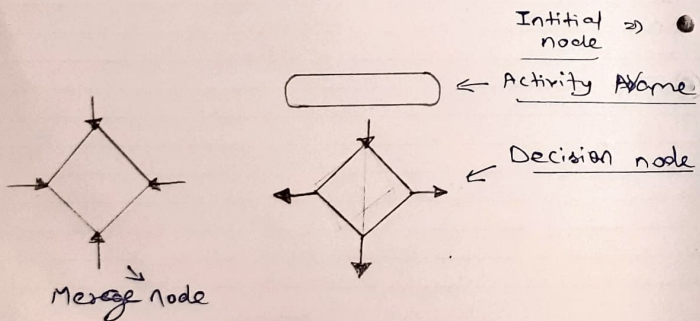
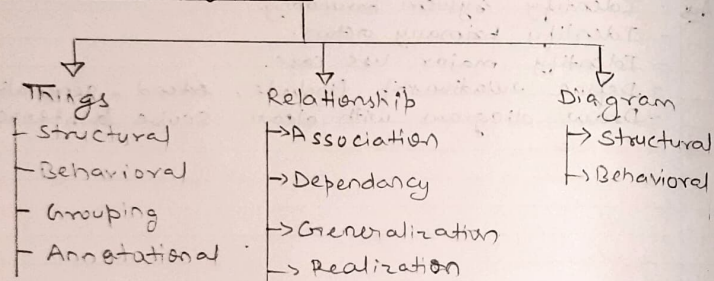


Basic Building Block -



Experiment - 4

Aim:- To understand & model the flow of activities in a system, using UML activity Diagram and state machine diagram.

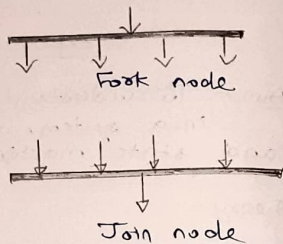
Theory -

Activity Diagram:-

- An Activity Diagram represent the workflow of a system, showing the sequence of activities & decision-making paths.
- It is mainly used for business process modeling & detailed logic of use cases.
- It is similar to a flowchart but supports parallel & concurrent processes.

(*) Building Blocks of Activity diagram:-

- Initial node - The starting point of the activity (black filled circle)
- Activity/Action - Represents a task or function (rounded rectangle)
- Decision Node - Represents branching with conditions (diamond)
- Merge Node - Combined multiple flows into one (diamond)



(Final node)



Swimlanes

Expt. No. _____

Date _____

Page No. 22

- (v) Fork node - Splits one flow into multiple parallel flows (thick horizontal bar)
(Branching)
- (vi) Join node - Synchronizes multiple parallel flows into one (thick horizontal bar)
(merge branching)
- (vii) Final node - End of the process (bullseye symbol)
- (ix) Swimlanes - Divide activities according to actors / roles.

State diagram:-

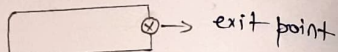
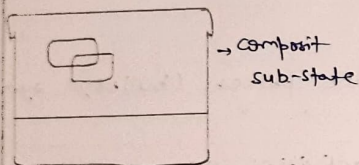
Procedure to draw an Activity Diagram:-

- (1) Identify the starting point (initial node)
- (2) List all the activities or actions in sequences.
- (3) Add decision points where branching occurs
- (4) Use fork / Join for parallel tasks if required.
- (5) Connect activities using control flows (arrow)
- (6) End with a final node.

Teacher's Signature _____

● → Initial node

⊙ → Final node



X → Terminate

→ ⇒ External Transition Arrows

* State Diagram:- A state diagram (also called state machine diagram) represents the states of an object & how it transitions from one state to another based on events.

It is useful for modeling reactive systems.
(Systems that respond to external events.)

Steps to draw a state diagram:-

- (1) Identify the object whose states need to be modeled.
- (2) Define all possible states of the object.
- (3) Identify the events the trigger transitions between states.
- (4) Mark the initial state (filled circle)
- (5) Draw transitions (arrow) between states with event labels.
- (6) Mark the final state (bullseye symbol)

Difference between Activity diagram & state diagram

Aspect	Activity Diagram	State Diagram
Purpose	Represent workflow or sequence of activities	Represent states of an object & transition.
Focus	Focuses on actions & control flow	focuses on object states & events.
Start/End	Starts with an initial node & ends with a final node.	Starts with an initial state & ends with a final state.
Use Case	Used for modeling business processes & system workflow	Used for modeling behaviour of Object in response to events
Nature	Similar to a flowchart	Similar to a finite state machine.

State diagram example -

States - New order → Processing → shipped → delivered

Transitions:-

Place order → New order,

Confirmation of Payment → Processing

Dispatch → Shipped

Customer Receives → Delivered.

Result:- The Concepts of Activity Diagrams & state diagrams were studied. The diagrams were successfully drawn for the sample system, & the difference between them were understood.

Teacher's Signature _____